4.5| Direct Map. 3 fields

1| A unique word / byte → word field

2| Tag field → which block is currently in cache.

3| line field → Where in the cache → specific memory block can be stored. (each block to only one permitted line).

⑥ Tag F , word F → Associative

⑦ Word F, Tag F,

Set Field; Divides the cache into sets. (each with multiple lines & slots).

tells in which set a mem block can be stored in. → data can occupy any line within that set.

⑧ temporal → recently used data & instrs

spatial → clustered mem location.

in cache memory & exploiting a cache hierarchy

use larger cache block + prefetching mechanism

(fetching items of anticipated use)

PROBLEMS

(1) $\frac{64 \text{ lines}}{4 \text{ sets}} = 16 \text{ sets} = 2^4 = 4$ bits needed to identify set number.

4K blocks $= 2^{12} \rightarrow (4096)$

128 words each.

$\therefore$ (set + tag) length = 12 bits $\therefore$ tag = 12 − 4 = 8 bit

128 words $= 2^7$. So 7 bits to specify each

	Tag	set	word
M.M Address:	8	4	7

line size, total

(2) (2) way set asso. 16 line byte, 8 KB size

64 MB mm. is byte addressable.

$$\frac{8\,Kb}{16\,byte} = 512 \text{ lines in cache.}$$

$\hookrightarrow 2^4$ = 4 bit → word

$$\text{total lines} = \frac{\text{cache size}}{\text{Line size}} = \frac{8192}{16} = 512$$

$$\therefore \frac{512 \text{ lines}}{2 \text{ lines/set}} = 256 \text{ set} = 2^8 = 8 \text{ bits}$$ (set)

Main mem size = 64 MB = 64×2^{20} byte

$= 2^{26} = 26$ bit

dividing by size of block,

$$\frac{64 \times 2^{20} \text{ bytes}}{16 \text{ bytes/block}} = 2^{22} \text{ blocks}$$

= set + tag

= 14 + 8 = 22

(26 − 8 + 4) = 14

tag	set	word
14	8	4

(3)

	Address	11 11 11	66 66 66	BB BB BB
Direct	Tag/Line/word	11/444/1	66/1999/2	BB/2EEE/3
Asso	Tag/word	4444/1	1999999/2	2EEEEE/3
Set Asso	Tag/set/word	22/444/1	cc/1999/2	177/EEE/3

111111 → 6

0001 0001 0001

1 2 3 4 5 6

8 12 4

↓ ↓ ↓

tag line word

(28) a) 1+4 = 5 clock cycles

b) 4 × (1+4) = 20 ll

c) miss penalty for 1 word + 3

= 8 cc.

		Line	
Direct:	8	14	2
Asso:	22		2
sel:	0	set 13	2

3 a) 111111 = 0001 0001 0001 0001 0001 0001 (2)

b) 666666 = 0110 0110 0110 0110 0110 0110 (2)

c) BBBBBB = 1011 1011 1011 1011 1011 1011 (2)

Direct Mapping:

a) Tag: 0001 0001 (11h)

Line: 00 0100 0100 0100 (0444h)

word: 01 (1h)

b) 666666

Tag: 0110 0110 (66h)

Line: 01 1001 1001 1001 (1999h)

Word: 10 (2h)

c) BBBBBB
Tag: 1011 1011 (BBh)

Line: 10 1110 1110 1110 (2EEEh)

word: 11 (3h)

Asso:

a) Tag: 00 0100 ... (044444h)

wor: 01 (1h)

b)

c)

④ @ Fig 4.10

Address length: 24bit → Tag + word + Line

of Addressable units: 2^{24}

block size: 4 → word field 2 bits → 2^2 = 4 words/block

of blocks in M.M: 2^{22} → $2^{24}/4 = 2^{22}$

of lines in cache: 2^{14} → Line field → 14 bits

size of tag: 8

tag field 8 bits.

Asso

→ 2^{22} →

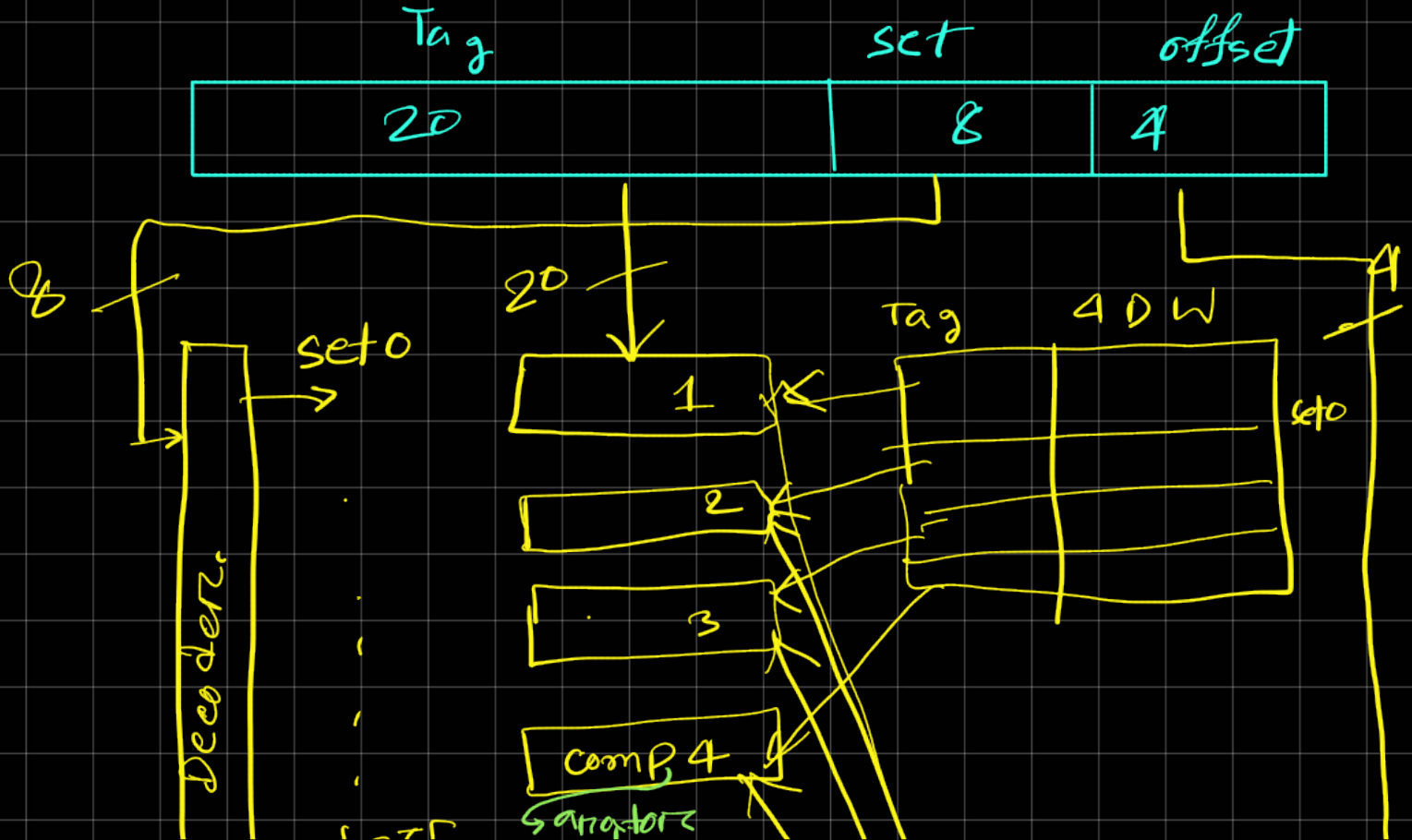
(5) 32bit mp., 16KB 4 way set asso cache. → lines.

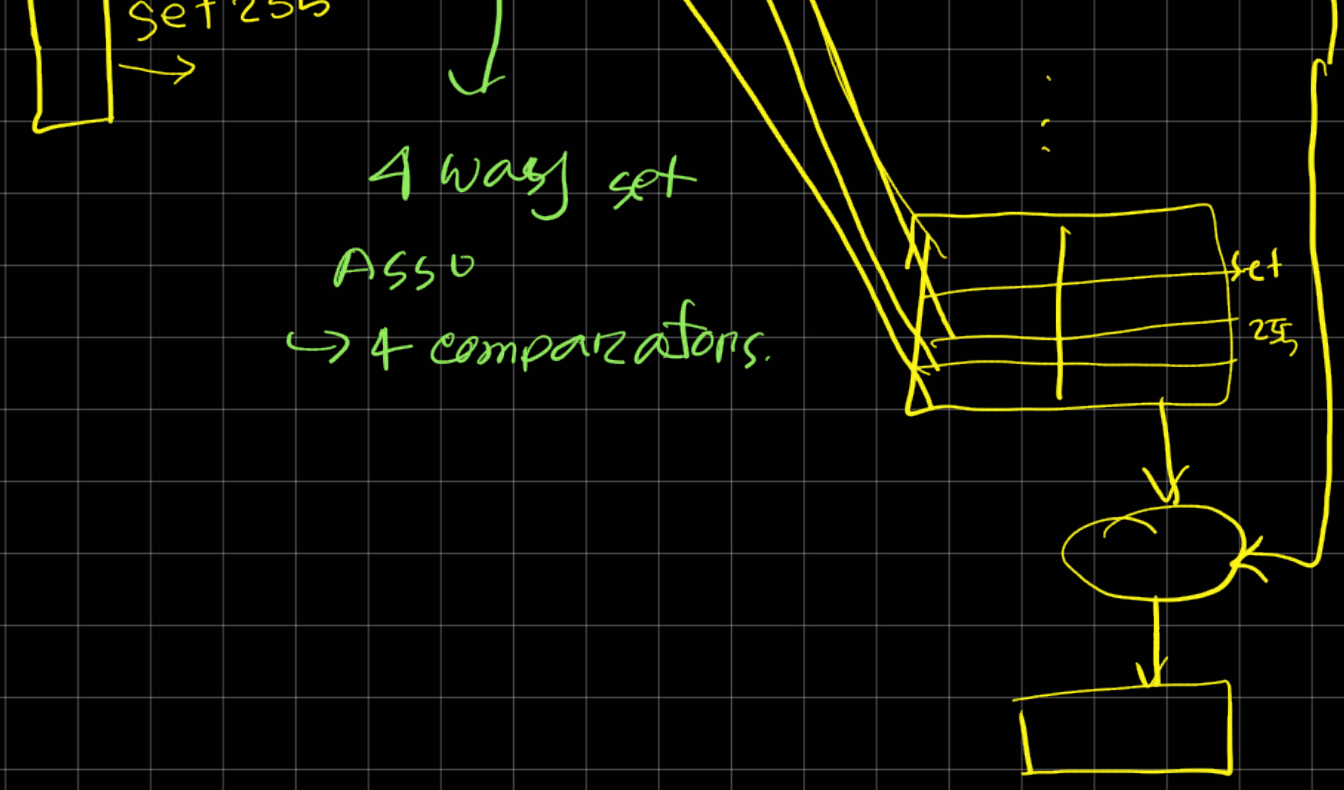
Line size → 4 [32] bit words. = 2^4 → 4 bytes = 16 bytes

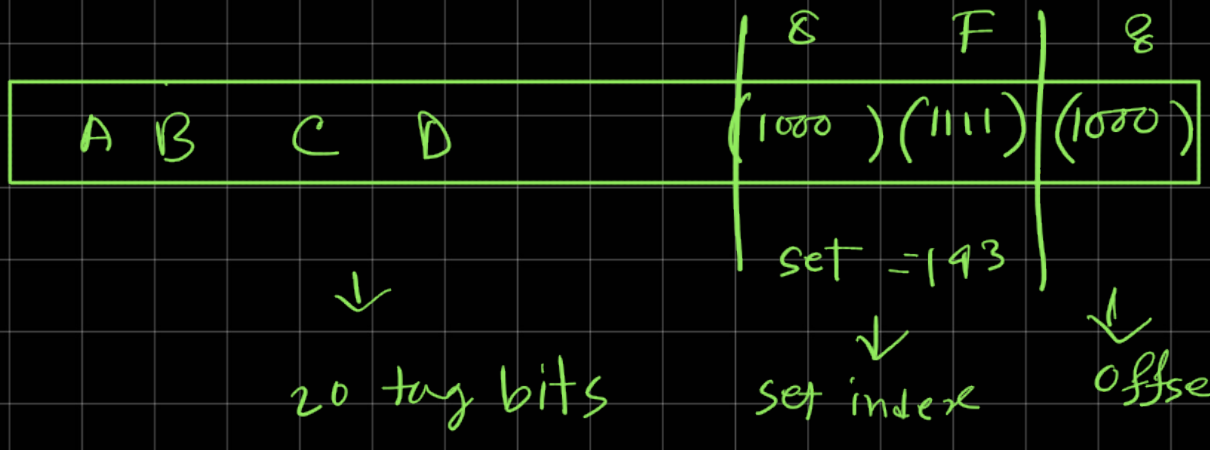
block frame size = 16 byte = 4 doublewords

of block " in cache = $\frac{16KB}{16B}$ = 1024 ✓

of sets = $\frac{\#\text{ Block frame}}{\text{Associativity}} = \frac{1024}{4} = 256$ sets $= 2^8$







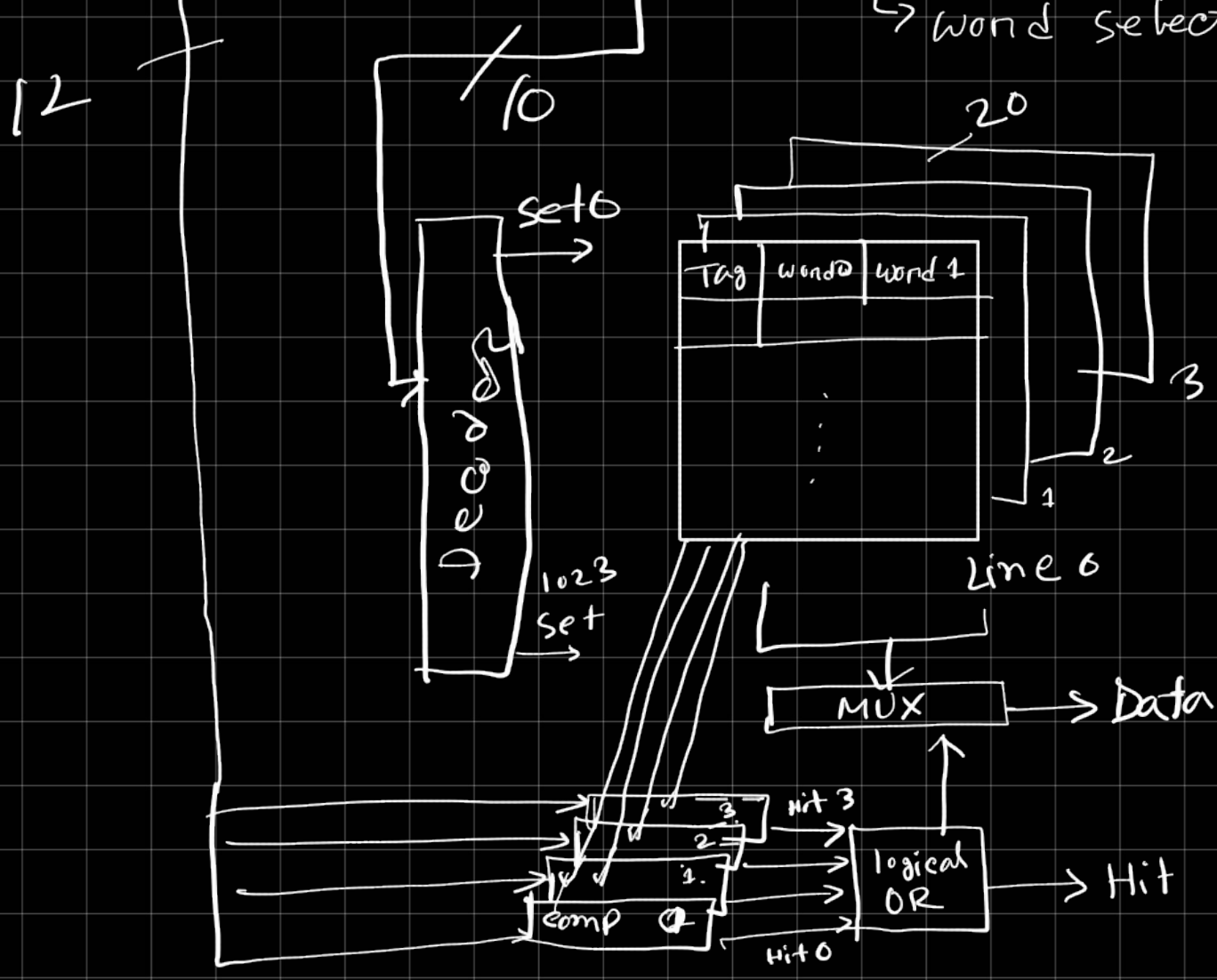
(6) 16 bit processor, 24 bit addresses.

4 way set associative

line size (16 bit words)

accomodate → 4K 32 bit words.





(7) 8KB cache, 4 way set Asso

block length 32 bit words. 128 sets.

1 line valid bit.

3 LRU bits (B_0, B_1, B_2)

On a cache miss, 80486 reads a 16 byte line from MM in a bus Mem read burst.

↳ 21 bit tag, 7 bit set ($128 = 2^7$), 4 bit word

fast set - 2 way, 4 times

Each set → 3 LRU, 4 lines

Each line → 4 32 bit words, a valid bit,
a 21 bit tag.

(8) (a) 16 bit Mem Addr to

Byte Number: byte offset (within the block)

↳ Block size = 8 bytes = 2^3 bytes

= 3 bits

Line Number: Number of cache line = 32 lines

$= 2^5$

= 5 bits

Tag: 16 - 3 - 5 = 8 bit

(b)

00011 → 3

⋮

10101 → 21

(c) 000 → 111

byte offset

(d) Total bytes of Mem stored in

cache = cache line × Block size per line

= 32 × 8 = 256

(e) Mem addr both point to line 3.

cache line contains tag.

if tag matches req add r → Cache hit

otherwise miss

(10) block size = 4 → 16 bit word → 2 byte = 8 byte per block.

Set size = 2 → Set associativity = K = 2

cache total words = 4096 → size = 4096 × 2 bytes = 8192 bytes

cachable M.M size = 64K × 32 bit

$\Rightarrow$ # of blocks in cache = $\frac{8192}{8}$ = 1024

of sets $S = \frac{C}{K} = \frac{1024}{2}$ = 512 sets

M.M size = 64K × 32 bit = 64K × 4 bytes

= 256 KB

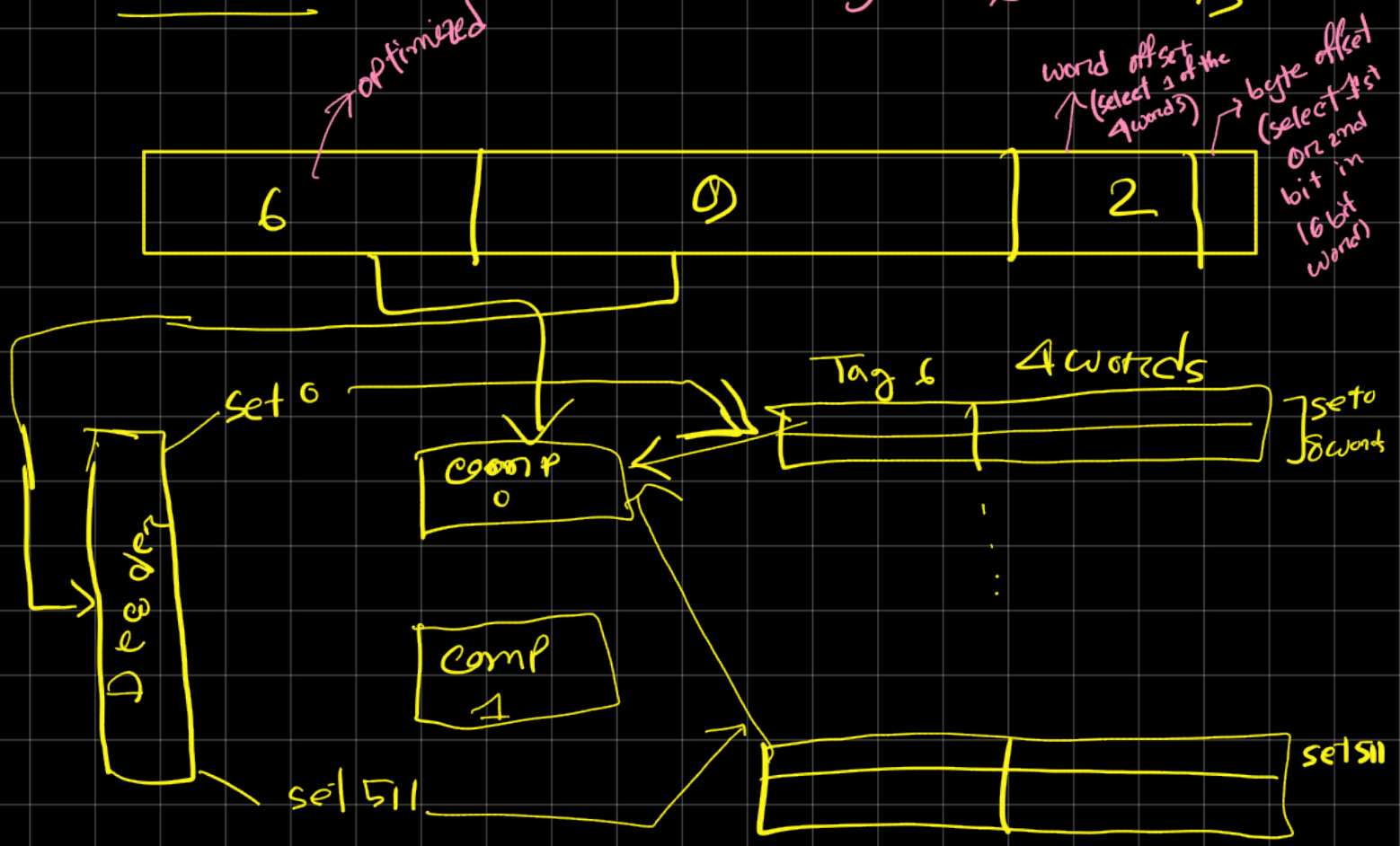
= 2^{18} = 18 bits

Block offset: 4 word per block

$= 8 \text{ bytes} = 2^3 \text{ bit} = 3 \text{ bits}$

set index: $512 = 2^9 = 9 \text{ bits}$

Tag : $18 - 9 - 3 = 6$ bits



4.17 M.M size = 32k 16 bit

$= 32k \times 2 \text{Byte} = 64 \text{KB}$

$= 2^{16} =$ 16 bit addr.

4K word cache 4 line sets → 64 words per line

(27) 2 level cache.

T_{c_1}

T_{c_2}

read operation,

$$T_a = [T_{c_1} + (1-H_1) T_{c_2}] + (1-H_2) T_m$$

(19) $T_{effective} = T_1 + (1-H) T_2$

↓ T_a ↓ cache ↓ mem

(26) (a) $T_a = T_c + (1-H) T_b + W(T_m - T_c)$

T_b → Time to transfer a line between cache & M.M.

W → fraction of write ref.

Write Through policy

(b) W_1 → probability → line is

cache altered.

Write back

$$T_a = T_e + (1-H)T_b + W_b(1-H)T_b$$

(25) (a) Mem cyc = 300 ns

IR process rate = 1 MIPS

$$= \frac{1}{10^6} = 1000 \text{ ns}$$

on Averge 2 bus.

∴ 600 ns

∴ 60%

(b) Hit ratio 0.5

instr fetch → 0.5 times

$= 0.5 \times 300 = 150 \text{ ns}$

operand access = 300 ns

tot = 450 ns. ∴ 0.45

(24) (A) $C_{c.} = \frac{1}{16.67 MHz} = 60ns$

$T_c = 2 \times 60 = 120ns$

$T_m = 3 \times 60 = 180ns$

$H = 0.9$

$T_a = (H \times T_c) + (1-H) \times T_m$

$= 126ns$

(6) 2 wait states add

$T_m' = (3+2) \times 60 = 300ns$

$T_a = (H \times T_c) + (1-H) \times T_m'$

$= 138ns.$

MM → 256

cache →

$2^8 \cdot 2^{20} = 2^{28}$

256 → 8 bit off

Cache → 512KB 2^{19}

$2^{28} / 2^{19} = 2^9$ tag bits